



A.M.($\leq$)	Any	Moon P.	Any/Bright
Image Quality		Seeing < 2.0"	
Cloudy Cover		Thin Cirrus (<70%)	

Title of Proposal: Multiband rotational photometry of Jupiter Trojans

Telescope 1.60m Perkin-Elmer

Abstract:

Jupiter Trojans are objects trapped in the L4 and L5 Lagrangian points of Jupiter. These objects were captured in this region during the scattering of bodies in the primordial Solar System; therefore, determining their physical properties and comparing them with other object classes is a fundamental step toward understanding their origin. Our proposal aims to perform multiband rotational observations of Trojan objects in both L4 and L5 to determine similarities among themselves and with other populations. This will provide information on rotation periods, magnitudes in each filter, and point-to-point color indices, in addition to an analysis of light curve morphology to investigate suppositions such as the occurrence of mutual phenomena with potential satellites or contact binarity. To achieve this, we propose observing these objects with the 1.60-m P&E telescope using SPARC4, which provides high image quality and simultaneous multiband observations.

Category: Solar System: Small Bodies (Asteroids, Comets, Moons, Kuiper Belt, Near Earth Objects)

Time requested: 112 hours.

Minimum time accepted: 56 hours.

Observing Mode: Classical Remote (CR)

Program Type: Regular Proposal (RP)

Principal Investigator (PI): Giuliano Margoti

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Is there a thesis or dissertation that will benefit from the data?: YES

Student	E-mail	Degree	Defense
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Giuliano Margoti	giulianomargoti@on.br	PhD student	11/2028
Eros De Oliveira Gradovski	erosgradovski@gmail.com	PhD student	03/2029

Optimal dates:

Between November 04 to February 26

Impossible dates:

NaN

Instruments:

SPARC4 (Photometry)

Detectors - 4 x iXon Ultra 888 CCDs

CCD Mode and Electron - Conventional Mode

Multiplying (EM) Gain

Scientific Justification

Studying small bodies is the key to understanding the dynamics of Solar System evolution. Studying different classes of objects and understanding the similarities between objects of the same class and how they resemble others is one of the ways to understand where these objects came from (SICARDY, B. et al. 2024). We currently understand that only three classes of objects are linked to the in situ formation of the Solar System: Main Belt asteroids, Kuiper Belt objects, and the Oort Cloud. Other classes of objects were, therefore, inserted into the regions where they are found today by external scattering processes (NESVORNÝ, D. et al. 2021).

The objective of this proposal is to study a class of objects known as Jupiter Trojans. These objects are trapped in Jupiter's L4 and L5 Lagrange points, and it is imagined that they were captured in this region after scattering caused by the instability in the Solar System due to Saturn passing through the 1:2 resonance with Jupiter (MORBIDELLI, A. et al. 2005). This pushed Uranus and Neptune into a massive asteroid belt, located between 25 and 35 astronomical units, scattering these objects and giving rise to, for example, the Centaur families, scattered TNOs, comets, and Jupiter Trojans (GOMES, R. et al. 2004, 2005).

Thus, understanding the physical properties of Jupiter Trojans is to look back at the young Solar System and how the objects within it formed, with what materials, colors, and angular momentum. These objects formed from a cloud of pebbles that had an angular momentum which would form different types of objects: objects with varied rotation periods, binary objects, objects with satellites, or contact binaries (NESVORNÝ, D. et al. 2021, JOHANSEN, A. et al. 2015). Due to close encounters with giant planets, objects with satellites or binaries underwent disruption or transformed into contact binaries (NESVORNÝ, D.; VOKROUHLICKÝ, D.. 2019). It is expected that a large portion of objects formed as binaries at the beginning of the Solar System (FRASER, W. C. et al. 2017); therefore, identifying objects in the Jupiter Trojan family that are contact binaries is one of the ways to verify this type of assumption.

A characteristic marked in objects that were once binaries is a slow rotation period due to possible tidal synchronization with their component; therefore, objects with very slow rotation are candidates for having been binaries at some point during their formation (COLLYER, C. et al. 2025). Furthermore, contact binaries have a signature in their rotation curves that reflects mutual occultations and/or eclipses between the components, which causes a signature in the rotation curve, for instance, a more rounded structure at light maxima while having a sudden "V-shaped" drop at minima (CARRY, B. et al. 2023). All these morphologies bring us clues as to where these objects came from and how they formed.

Knowing the colors of these objects is also a way to determine if they belong to the same formation family. It is expected, therefore, that if they do, they have similar color indices and do not have very large albedo variations given their rotation (THIROUIN, A.; SHEPPARD, S. S. 2018). Until now, we only have rotation curves made in a specific filter, such as the Sloan R filter, or in Clear for these objects; therefore, it is impossible to say whether the light curve morphology is caused by the object's shape variation or albedo. This makes it difficult to determine, for example, the object's shape, since we believe that the brightness variation is always entirely caused by shape variation (Margoti, G. et al. 2026). Knowing if the object has albedo variation, therefore, brings more accuracy to the determination of the object's shape.

Depending on the time of year, we can better observe objects in L4 or L5. In 2026B. Our target list contains objects of this class that will be above 35 degrees for more than 5 hours throughout the night during the period of this request, and they are complementary, meaning that in both parts of the night, we have visible targets.

Given everything discussed here, our proposal is to use the SPARC4's capability to observe in four filters simultaneously to obtain the multiband synchronous rotation of our targets, thus being able to state whether the object's variation is caused entirely by shape or if significant albedo variation exists on the body, in addition to being able to analyze the morphology of the light curves to classify some targets as possible contact binaries. To do this, we will use a proprietary reduction and analysis pipeline, which includes, for example, the photometry part, Lomb-Scargle periodogram analysis, and light curve analysis using three-dimensional models. and we intend to compare our data with the results from the SPARC4 reduction pipeline.

Experimental Design

Some of the targets in our campaign have stellar occultation data with peculiar results, such as the case of Cebriones, Paumys, Hektor (figure 1), Mentor and Thymbraeus (figure 2), where in one occultation we detected the disappearance and reappearance of the star twice, indicating that the object has large topography variations, is a contact binary object or have a satellite. Mentor and Thymbraeus were our targets in a campaign conducted in 2025B for multiband rotation observation (proposal P-005). In Thymbraeus, we detected the characteristics in the light curve morphology to classify it as a contact binary (Figure 3), and for Mentor, an albedo variation (Figure 4).

Our targets were selected to obtain light curves with high photometric quality. Required exposure times range between 60 and 180 seconds; therefore, very faint objects are not feasible. The rotation periods could not be synchronous with Earth's rotation (i.e., an integer fraction of Earth's day), and the rotational amplitude must be greater than 0.1 magnitudes. Among the objects we study, those that meet these characteristics are the Jupiter Trojans.

Jupiter Trojans are classified into two families (Greeks and Trojans) located at Jupiter's L4 and L5 Lagrange points. The rotation period of our targets is varied, ranging from approximately 4h to 65h, with the majority between 7 and 10 hours. The rotational amplitudes vary between 0.1 and 0.3 magnitudes, and the visual magnitudes vary between 15.2 and 18.6, being compatible with the observation needs of the OPD 1.60-m (P&E) telescope to obtain a signal to noise of 20.

Our intention is to study objects in both Jupiter's L4 and L5 to verify their similarities in the properties mentioned previously. Therefore, this project is configured as a Large Program, which we intend to submit in subsequent semesters. Each set of objects will provide us with data for isolated or joint publication, depending on the quality and coverage of the light curves.

The required period for our observations spans from November 15th to February, with the most favorable dates occurring in February. To detect a significant number of objects, we request a minimum of 7 observing nights, averaging 8 hours per night (totaling 56 hours). For optimal results, however, we request 14 nights (averaging 8 hours per night, totaling 112 hours). Each selected target will be observed for as long as possible, taking into account the rotation periods reported in the literature.

We request that the nights be distributed in sequences of at least 3 consecutive nights to allow for consistent tracking of the targets. Ideally, the preferred strategy is to have the largest possible number of consecutive nights split into two observing runs, for example, 7 nights in December and 7 nights in February. This approach also helps mitigate potential losses due to weather conditions and target visibility.

Our target list includes objects in both L4 and L5. A selection of these targets for February 15th is shown in Table 1. This table is ordered by magnitude and presents the objects' elevation in degrees for the indicated hours. This is only a subset of our targets; other objects will be visible during that night, in addition to those visible until December. Observing dates in December and February are preferred precisely because we have targets that cover the entire night, whereas in November, most targets are visible for only 4/5 of the night.

Our intention with these data is the publication of rotational light curves, with an analysis of periods, colors, and color variations as a function of rotation. To achieve this, we will use our proprietary photometry and data analysis pipeline, and we also intend to compare our results with those obtained from the SPARC4 reduction pipeline. And of course, we will perform the appropriate image calibrations for each night of observation.

To obtain the periodogram, we will use the Lomb-Scargle technique, with rotation period uncertainties calculated using the methodology developed by Gradoviski et al. (2026). If shape models for these objects exist in the literature, we will test them to verify their quality. If they are found to be inadequate, we intend to use our data in future works, in combination with stellar occultations, to determine new or refined three-dimensional shape models using the methodologies developed by Margoti et al. (2026) and Gradoviski et al. (2026).

Consequently, our main observables are object magnitude versus time, from which we will derive the rotation period and phase of maximum brightness. We will also produce point-to-point color indices and provide the necessary data for future 3D shape reconstruction.

Technical Description

For this proposal, the required instrument suite is the 1.60-m (P&E) telescope and the SPARC4 camera, with no requirement for additional filters. The use of SPARC4 is necessary as this project aims to perform simultaneous multiband rotational photometry of Trojan objects, a methodology not yet implemented for this class. The P&E telescope is essential to achieve the expected photometric precision with a 60-second cadence and because it serves as the host for SPARC4.

The proposed observations do not require a specific Moon phase; depending on the angular separation between the target and the Moon, comparable results can be obtained. However, cloud cover is a critical factor. High optical depth cloud cover during observations may compromise the scientific goals. Rapid passages of thin cirrus are less detrimental to the photometric data.

Airmass constraints are flexible; although lower airmass optimizes the signal, the strategy involves monitoring the targets for the maximum possible duration, observing down to the telescope's tilt limits. If primary target coverage is optimized, additional targets will be observed for 2 to 3 hours at elevations above 50 degrees.

An SNR ≥ 20 is required to ensure high-quality results, making sky condition requirements flexible. Exposure times will range from 60 to 200 seconds. Based on the SPARC4 ETC calculations, our brightest target (624 Hektor, mag 15.1) yields an SNR of 128, 201, 165, and 121 in the g' , r' , i' , z' channels, respectively, with a 60-second exposure. For the faintest target (20729 Opheltius, mag 18.5), a 200-second exposure results in an SNR of 20, 33, 30, and 20 in the respective channels. These estimates assume standard configurations: Quarter Moon, regular image quality, airmass = 2, 0.1 MHz acquisition mode with gain 2, and a 4-pixel photometric aperture.

To obtain multiband rotation curves for a minimum of 5 objects, we request 7 observing nights (averaging 8h/night, totaling 56 hours). To enhance the statistical significance for publication, we aim to observe at least 10 objects, requiring a total of 14 nights (112 hours). A subset of our targets is listed in Table 1, which presents their apparent magnitudes for February, their rotation periods, and their elevations throughout the night. This subset represents only a fraction of our full target list, which contains 100 Trojan objects visible from mid-November through February.

Figures and Tables

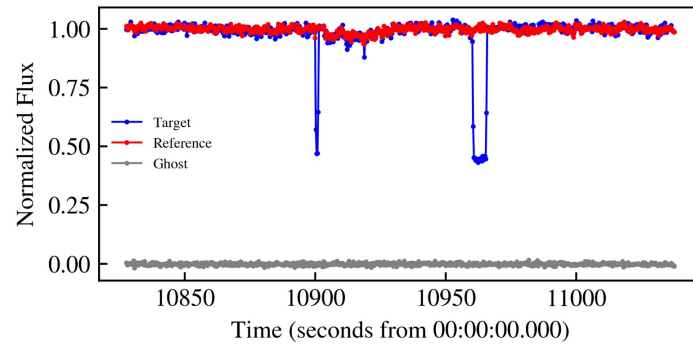


Figure 1: Stellar occultation by Hektor observed in SOAR, where we detected a drop in the star's brightness before the passage of the main body, which, given the distance, preferentially indicates a satellite. Since we did not detect a contact binary signature in its light curves, this is the most accepted assumption. Credits: Eros O. Gradovski

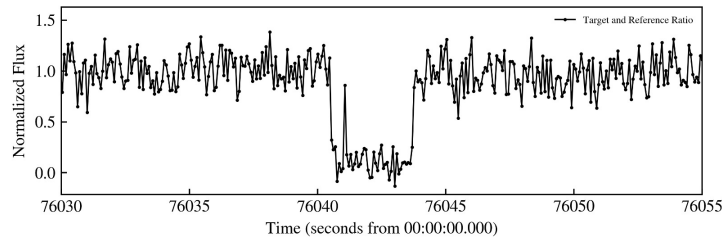


Figure 2: Occultation light curve of Thymbraeus; in this curve, there was a sudden reappearance of the star during the passage of the main body, indicating considerable relief in a region of the body. As the light curves of Thymbraeus show contact binary characteristics, we understand this reappearance to be the separation of the components in the sky plane at that instant. Credits: Eros O. Gradovski

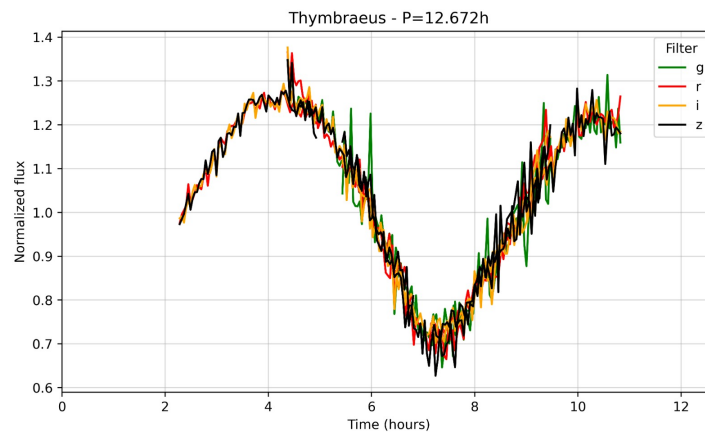


Figure 3: Rotational light curve of Thymbraeus obtained over 4 nights of observation at the OPD 1.60-m telescope with SPARC4. We see in this rotation curve the characteristics for the object to be a contact binary, such as a sudden "V-shaped" drop at the brightness minimum and a more rounded shape at the brightness maxima. This characteristic, combined with the occultation observations that show a large topographic signal, are factors that strengthen the idea of Thymbraeus being a contact binary. Credits: Eros O. Gradovski

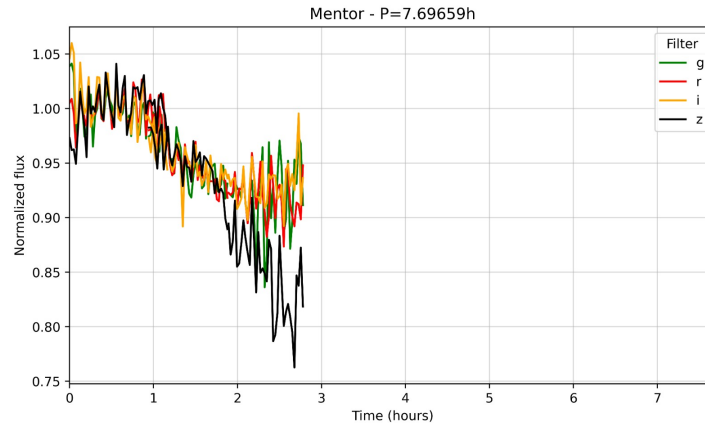


Figure 4: Light curve of Mentor obtained over 3 days of observation at the OPD 1.60-m telescope with SPARC4. Rotation data for Mentor were acquired 1h in each night, in which we detected an atypical flux variation only in the SPARC4 z' filter, possibly indicating that Mentor has a redder region near a brightness minimum. Credits: Eros O. Gradvoski

Table 1: Elevation of Jupiter Trojan targets during the February 15 at OPD. The columns indicate the object name, the magnitude on the specific date, the rotation period in hours available in the Small Body Database (SBDB), and the hours during the night, indicating the object's elevation in degrees at each time.

Objeto	mag	rot (h)	18:00	19:00	20:00	21:00	22:00	23:00	00:00	01:00	02:00	03:00	04:00	05:00	06:00
624 Hektor (A907 CF)	15.1	6.9	—	—	—	—	—	—	—	—	44.1	57.3	70.5	82.5	79.9
911 Agamemnon (A919 FB)	15.9	6.6	—	—	—	—	—	—	—	39.5	52.8	66.2	79.2	84.0	71.9
3451 Mentor (1984 HA1)	16.0	7.7	—	46.1	58.8	69.0	71.8	64.6	52.8	39.7	—	—	—	—	—
1172 Aeneas (1930 UA)	16.0	8.7	35.4	44.6	50.5	51.2	46.7	38.2	—	—	—	—	—	—	—
1143 Odysseus (1930 BH)	16.1	10.1	—	—	—	—	—	—	—	—	46.5	60.2	73.0	80.0	72.2
4709 Ennomos (1988 TU2)	16.1	12.3	36.8	46.2	52.0	52.6	47.7	38.9	—	—	—	—	—	—	—
2207 Antenor (1977 QH1)	16.3	8.0	36.0	45.1	50.7	51.1	46.3	37.6	—	—	—	—	—	—	—
588 Achilles (A906 DN)	16.4	7.3	—	—	—	—	—	—	—	—	35.3	48.5	61.9	75.1	85.3
3317 Paris (1984 KF)	16.4	7.1	46.1	57.9	66.4	67.3	60.1	48.7	35.8	—	—	—	—	—	—
659 Nestor (A908 FE)	16.6	16.0	—	—	—	—	—	—	—	43.0	56.3	68.2	74.8	69.8	58.3
2363 Cebriones (1977 TJ3)	16.7	20.1	49.7	62.7	73.3	74.7	65.5	52.7	39.2	—	—	—	—	—	—
3709 Polypoites (1985 TL3)	16.9	10.0	—	—	—	—	—	—	—	37.5	51.0	63.6	72.9	72.4	62.7
15436 Dexius (1998 VU30)	17.0	9.0	—	—	—	—	—	—	—	36.6	50.0	63.6	77.2	87.0	74.8
3540 Protesilaos (1973 UF5)	17.0	8.9	—	—	—	—	—	—	—	—	35.3	49.2	63.1	76.8	86.1
4035 Thestor (1986 WD)	17.4	13.5	—	—	—	—	—	—	—	40.8	54.7	68.5	81.5	81.0	67.9
4722 Agelaos (4271 T-3)	17.4	18.4	35.1	40.7	42.3	39.5	—	—	—	—	—	—	—	—	—
2895 Memnon (1981 AE1)	17.4	7.5	51.7	62.7	68.8	65.9	56.3	44.0	—	—	—	—	—	—	—
4754 Panthoos (5010 T-3)	17.6	27.7	40.1	50.0	56.2	56.5	50.8	41.2	—	—	—	—	—	—	—
38050 Bias (1998 VR38)	17.8	18.9	—	—	—	—	—	—	—	42.4	55.6	68.8	80.9	81.0	68.9
21900 Orus (1999 VQ10)	17.8	13.4	—	—	—	—	—	—	—	—	46.9	60.8	74.7	87.0	76.7
11395 Iphinous (1998 XN77)	17.8	13.7	—	—	—	—	—	—	—	42.7	56.6	70.4	83.2	79.7	66.2
12929 Periboea (1999 TZ1)	17.8	9.3	54.8	68.5	82.4	83.7	69.9	56.1	42.4	—	—	—	—	—	—
5258 Rhoeo (1989 AU1)	17.9	19.9	—	—	—	—	—	—	—	37.0	50.9	64.6	77.5	81.6	70.2
14268 (2000 AK156)	18.4	7.5	—	—	—	—	—	—	38.8	52.5	66.4	80.2	85.6	71.8	57.9
20729 Opheltius (1999 XS143)	18.5	5.7	—	—	—	—	—	—	—	42.4	55.0	67.2	77.5	77.9	67.9

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Previous missions of this proposal to this telescope

A proposal similar to this one was submitted in the 2025B for the observation of a Centaur and a Trojan (P-005). The primary target was the Centaur Chariklo, which had an apparent magnitude of 19 at the time; however, Chariklo's rotational amplitude was expected to be less than 0.05. We acquired imaging data for Chariklo's rotation over 5 days, also taking advantage of one night from another proposal (P-008) by the same author. In this second proposal, the request was for only 1 hour per night, with the observation of Chariklo serving as backup. Nevertheless, these results for Chariklo were not sufficient to reach the required conclusions; our light curve did not achieve enough quality to constrain Chariklo's amplitude. Consequently, these data served to establish a minimum amplitude of less than 0.08 and a global color for Chariklo, which is consistent with the literature.

In the same observing run, we also observed the Trojan Thymbraeus (Figure 3) as a secondary target, achieving a coverage of slightly more than half a rotational phase. We are currently working on the point-to-point color determination for this object. This observation served as a starting point for the present proposal, demonstrating that it is possible to observe these objects with adequate precision, and it also served as the test case for our reduction pipeline. During this project, we also observed approximately 1/4 of the Trojan Mentor's rotation (Figure 4), in which we noted an atypical albedo variation in the z' filter; this finding further motivated the submission of the current time propose.

Previous missions of other proposals to this telescope

In 2025A and 2025B, the author of this proposal submitted two other time requests for the P&E telescope using SPARC4. The proposals were P-008 and P-008, respectively. Two projects with the same objective: observing the mutual phenomena of Saturn's satellites, which occur every 14 years. While this is not the primary author's main line of research, he was responsible for executing these proposals. We successfully utilized 60% of the allocated nights, with losses attributed to both technical issues with SPARC4 and weather conditions.

The analysis of these images is currently underway by the Brazilian part of international community as part of a global data collection effort. In the observations successfully executed, we achieved a 33% efficiency rate; losses were primarily caused by the satellites' proximity to Saturn, which led to flux contamination from either the planet or its rings. A significant cause of this difficulty was the absence of a methane filter for the observations, which would have significantly reduced Saturn's brightness contribution in the images. For this request, the P&E telescope was necessary because the targets had magnitudes greater than 12 and required a high cadence, as some events were of short duration with very shallow magnitude drops.

Since the time requirement for those nights was approximately 2 hours, our backup for these projects was the astrometric observation of TNOs and Centaurs. This provided astrometric positions that were added to our databases, allowing us to improve the ephemerides of 12 objects. Furthermore, we are currently processing these data through our reduction pipeline to obtain colors for these objects for potential future publications.

Previous results in the field by the Principal Investigator

The Master's thesis of the PI of this proposal was focused on the determination of the physical properties of Quaoar and Chariklo, utilizing stellar occultation and rotational data "Determinação da Forma Tridimensional de (50000) Quaoar por Ocultações Estelares Utilizando Algoritmo Genético para Minimização". Upon entering the PhD program, the PI's first first-authored paper was written, presenting the developed methodology and the results for Quaoar "Size, shape, density, and atmospheric limit of (50000) Quaoar revealed from 14 years of stellar occultation"; this manuscript is currently under review in The Astrophysical Journal (ApJ).

The PI, together with the co-PIs of this proposal, Eros Gradovski and Felipe Braga Ribas, developed a technique for analyzing occultations in conjunction with rotational light curves to determine the shape models of Trojans, which is also being applied to Centaurs. At this stage, this proposal is essential for taking a step forward in studying the colors and albedo variations of these objects.

Among the papers co-authored by the PI are:

- Evidence of a New Arc or a Belt of Small Satellites around (50000) Quaoar

- Constraining the size, shape, and albedo of the large trans-Neptunian object (28978) Ixion with multi-chord stellar occultations
- The rings of (2060) Chiron: Evidence of an evolving system
- Nsf-doe vera c. rubin observatory observations of interstellar comet 3i/atlas (c/2025 n1)
- Investigating the formation of small Solar System objects using stellar occultations by satellites: present, future and its use to update satellite orbits
- Physical characteristics of Jupiter's Trojan (1437) Diomedes from a tri-chord stellar occultation in 2020 and dimensionless three-dimensional model
- Kilometer-precise (Ull) Umbriel physical properties from the multichord stellar occultation on 2020 September 21
- A large topographic feature on the surface of the trans-Neptunian object (307261) 2002 MS4 measured from stellar occultations
- Constraints on (2060) Chiron's size, shape, and surrounding material from the November 2018 and September 2019 stellar occultations
- The two rings of (50000) Quaoar
- A dense ring of the trans-Neptunian object Quaoar outside its Roche limit
- A multi-chord stellar occultation by the large trans-Neptunian object (174567) Varda

The PI of this proposal also actively participates in national and international events to present his results. The most recent ones are:

- IAU Symposium 401: Advancing Reference Systems, Ephemeris, and Standards (2025): Three-Dimensional Shape of (50000) Quaoar and Orbit Refinement Using Data from 29 Stellar Occultations
- XLVIII Reunião anual da sociedade astronômica Brasileira (2025): Modeling the Physical Properties of Outer Solar System Objects Using Genetic Algorithms and Multi-technique Observations
- 35° Colóquio Brasileiro de Matemática (2025): Modeling Physical Properties of Outer Solar System Objects.
- XXII Congresso Brasileiro de Dinâmica Orbital e Planetologia (2024): Equilibrium Figures for Asteroids.

PI and Co-Is publications relevant to this proposal

The co-authors of this proposal have extensive experience with rotational light curves, such as the Master's thesis presented in 2025 by Eros de Oliveira Gradovisky. His work utilized rotational and occultation data to develop irregular, concave shape models for several Jupiter Trojans: "Obtenção dos Parâmetros Rotacionais e Escala de Troianos de Júpiter Através de Curvas de Rotação e Ocultações". Furthermore, our team has produced numerous studies focused on determining the physical properties of Trojans, Centaurs, and Trans-Neptunian Objects (TNOs), the main ones are:

- A ring system detected around the Centaur (10199) Chariklo
- Size and shape of Chariklo from multi-epoch stellar occultations
- Refined physical parameters for Chariklo's body and rings from stellar occultations observed between 2013 and 2020
- The size, shape, density and ring of the dwarf planet Haumea from a stellar occultation
- A Pluto-like radius and a high albedo for the dwarf planet Eris from an occultation
- Albedo and atmospheric constraints of dwarf planet Makemake from a stellar occultation
- Triplicity and physical characteristics of Asteroid (216) Kleopatra
- The size, shape, albedo, density, and atmospheric limit of transneptunian object (50000) Quaoar from multi-chord stellar occultations

- Study of the Plutino Object (208996) 2003 AZ84 from Stellar Occultations: Size, Shape, and Topographic Features
- Changing material around (2060) Chiron revealed by an occultation on December 15, 2022
- Physical characterization of double asteroid (617) Patroclus from 2007/2012 mutual events observations
- The large trans-Neptunian object 2002 TC302 from combined stellar occultation, photometry, and astrometry data
- Results from a triple chord stellar occultation and far-infrared photometry of the trans-Neptunian object (229762) 2007 UK126
- Physical properties of the trans-Neptunian object (38628) Huya from a multi-chord stellar occultation
- A stellar occultation by the transneptunian object (50000) Quaoar observed by CHEOPS
- Results of two multichord stellar occultations by dwarf planet (1) Ceres
- The Trans-Neptunian Object (84922) 2003 VS2 through Stellar Occultations
- The multichord stellar occultation on 2019 October 22 by the trans-Neptunian object (84922) 2003 VS2
- The 2017 May 20 stellar occultation by the elongated centaur (95626) 2002 GZ32
- Physical properties of Centaur (60558) 174P/Echeclus from stellar occultations

The co-PIs of this proposal have also presented relevant works on the study of Trojan objects at national and international events; the primary ones are:

- XLVIII Reunião Anual da Sociedade Astronômica Brasileira (2025). Obtenção da forma tridimensional do Troiano (911) Agamemnon através de curvas de rotação e ocultações estelares.
- XLVII Reunião Anual da Sociedade Astronômica Brasileira (2024). Obtaining (911) Agamemnon size and rotational parameters through its 3D model and new Stellar Occultations.
- XII Taller de Ciencias Planetarias (2024). Studying Jupiter Trojans through stellar occultations.
- XXII Colóquio Brasileiro de Dinâmica Orbital (2024). Testing (911) Agamemnon 3D model with new Stellar Occultations and synthetic light curves generations.
- II Workshop de Astronomia e Astrofísica do Paraná (2023). Ocultações estelares por (3451) Mentor, um troiano possivelmente duplo.
- XLVI Reunião Anual da Sociedade Astronômica Brasileira (2023). Ocultações estelares por (624) Hektor, o maior troiano conhecido.

Objects List.

Object	A.R.(h,m)	Dec.(deg)	Mag.	Band	Type	Comments
588 Achilles (A906 DN)	13:00:47.87	-16:20:38.29	15.9	V	Sci	position at 2026-02-01
617 Patroclus (A906 UL)	2:32:44.70	19:06:06.70	15.7	V	Sci	position at 2026-02-01
624 Hektor (A907 CF)	12:19:02.12	-8:20:42.66	14.8	V	Sci	position at 2026-02-01
659 Nestor (A908 FE)	11:14:10.96	5:50:47.79	16.4	V	Sci	position at 2026-02-01
884 Priamus (A917 SU)	3:17:30.10	26:26:10.36	16.7	V	Sci	position at 2026-02-01
911 Agamemnon (A919 FB)	11:57:48.91	-7:47:08.49	15.5	V	Sci	position at 2026-02-01
1143 Odysseus (1930 BH)	11:38:25.72	-0:54:48.81	15.6	V	Sci	position at 2026-02-01
1172 Aeneas (1930 UA)	3:27:28.22	19:43:45.85	16.1	V	Sci	position at 2026-02-01
1173 Anchises (1930 UB)	1:25:45.66	15:38:31.90	16.6	V	Sci	position at 2026-02-01
1208 Troilus (1931 YA)	2:48:04.97	22:38:44.02	16.7	V	Sci	position at 2026-02-01
1404 Ajax (1936 QW)	11:14:17.25	0:43:15.90	16.9	V	Sci	position at 2026-02-01
1437 Diomedes (1937 PB)	13:35:12.78	-31:13:05.93	16.0	V	Sci	position at 2026-02-01
1583 Antilochus (1950 SA)	13:14:16.04	-23:14:54.68	16.4	V	Sci	position at 2026-02-01
1749 Telamon (1949 SB)	12:26:53.71	-5:14:52.38	16.8	V	Sci	position at 2026-02-01
1867 Deiphobus (1971 EA)	2:42:43.06	36:38:43.42	16.1	V	Sci	position at 2026-02-01
1868 Thersites (2008 P-L)	11:05:28.27	-6:11:23.44	17.3	V	Sci	position at 2026-02-01
1873 Agenor (1971 FH)	1:19:24.82	3:48:44.03	18.4	V	Sci	position at 2026-02-01
2146 Stentor (1976 UQ)	12:23:16.34	28:47:31.42	17.7	V	Sci	position at 2026-02-01
2207 Antenor (1977 QH1)	3:05:49.13	10:14:31.04	16.6	V	Sci	position at 2026-02-01
2223 Sarpedon (1977 TL3)	2:56:32.69	12:12:31.39	17.0	V	Sci	position at 2026-02-01
2241 Alcathous (1979 WM)	1:07:05.87	21:18:40.53	16.7	V	Sci	position at 2026-02-01
2260 Neoptolemus (1975 WM1)	12:39:50.21	17:53:30.19	17.0	V	Sci	position at 2026-02-01
2357 Phereclos (1981 AC)	2:43:58.45	13:33:54.69	17.0	V	Sci	position at 2026-02-01
2363 Cebriones (1977 TJ3)	3:30:12.70	0:30:19.57	16.9	V	Sci	position at 2026-02-01
2456 Palamedes (1966 BA1)	12:17:53.03	-11:10:04.44	16.8	V	Sci	position at 2026-02-01
2674 Pandarus (1982 BC3)	2:29:09.49	13:12:41.58	17.2	V	Sci	position at 2026-02-01
2759 Idomeneus (1980 GC)	12:25:52.44	-0:07:58.92	17.2	V	Sci	position at 2026-02-01
2797 Teucer (1981 LK)	13:50:11.56	11:49:23.92	16.2	V	Sci	position at 2026-02-01
2893 Peiroos (1975 QD)	2:28:00.44	0:50:01.68	16.9	V	Sci	position at 2026-02-01
2895 Memnon (1981 AE1)	2:47:34.74	-12:22:37.11	17.7	V	Sci	position at 2026-02-01
2920 Automedon (1981 JR)	12:04:19.62	-21:04:19.24	16.3	V	Sci	position at 2026-02-01
3063 Makhaon (1983 PV)	12:32:10.23	-17:51:32.47	16.0	V	Sci	position at 2026-02-01
3240 Laocoon (1978 VG6)	2:15:57.14	15:55:48.13	18.1	V	Sci	position at 2026-02-01
3317 Paris (1984 KF)	3:42:45.16	-9:12:39.64	16.5	V	Sci	position at 2026-02-01
3451 Mentor (1984 HA1)	4:30:20.90	-4:38:34.85	16.2	V	Sci	position at 2026-02-01
3540 Protesilaos (1973 UF5)	12:59:16.80	6:49:43.47	16.5	V	Sci	position at 2026-02-01
3548 Eurybates (1973 SO)	13:44:34.68	-6:37:15.89	17.7	V	Sci	position at 2026-02-01
3564 Talthybius (1985 TC1)	11:09:07.87	20:59:11.68	16.6	V	Sci	position at 2026-02-01
3596 Meriones (1985 VO)	13:03:27.44	-13:30:33.12	17.0	V	Sci	position at 2026-02-01
3708 Socus (1974 FV1)	2:52:55.04	27:53:48.84	16.7	V	Sci	position at 2026-02-01
3709 Polypoites (1985 TL3)	11:02:51.22	-4:55:08.10	16.4	V	Sci	position at 2026-02-01
3793 Leonteus (1985 TE3)	12:54:30.19	-11:55:32.51	16.3	V	Sci	position at 2026-02-01
4035 Thestor (1986 WD)	11:10:43.46	-9:09:19.97	16.9	V	Sci	position at 2026-02-01
4060 Deipylus (1987 YT1)	12:09:48.05	0:37:27.03	16.4	V	Sci	position at 2026-02-01
4063 Euforbo (1989 CG2)	13:38:49.46	11:16:13.54	16.5	V	Sci	position at 2026-02-01
4068 Menestheus (1973 SW)	11:25:26.14	-1:28:24.42	16.6	V	Sci	position at 2026-02-01
4086 Podalirius (1985 VK2)	13:46:38.51	6:14:46.50	17.3	V	Sci	position at 2026-02-01
4348 Poulydamas (1988 RU)	1:56:16.20	11:29:36.27	17.2	V	Sci	position at 2026-02-01

4489 Dracius (1988 AK)	11:13:57.45	31:58:07.51	16.1	V	Sci	position at 2026-02-01
4543 Phoinix (1989 CQ1)	13:30:56.37	-23:07:11.02	17.7	V	Sci	position at 2026-02-01
4708 Polydoros (1988 RT)	0:42:40.56	11:27:29.65	17.8	V	Sci	position at 2026-02-01
4709 Ennomos (1988 TU2)	3:12:47.31	23:14:27.89	16.4	V	Sci	position at 2026-02-01
4715 Medesicaste (1989 TS1)	2:29:53.82	31:44:08.63	17.4	V	Sci	position at 2026-02-01
4722 Agelaos (4271 T-3)	2:11:03.19	10:10:57.20	17.5	V	Sci	position at 2026-02-01
4754 Panthoos (5010 T-3)	3:14:06.28	4:50:14.34	17.9	V	Sci	position at 2026-02-01
4792 Lykaon (1988 RK1)	2:06:35.70	9:48:29.95	17.9	V	Sci	position at 2026-02-01
4805 Asteropaos (1990 VH7)	1:21:43.55	19:32:55.56	17.7	V	Sci	position at 2026-02-01
4832 Palinurus (1988 TU1)	3:24:48.62	-1:14:44.44	17.6	V	Sci	position at 2026-02-01
4833 Meges (1989 AL2)	11:55:39.54	38:30:04.61	16.7	V	Sci	position at 2026-02-01
4834 Thoas (1989 AM2)	13:04:54.98	28:24:52.09	16.8	V	Sci	position at 2026-02-01
4836 Medon (1989 CK1)	13:59:16.76	8:51:34.16	17.1	V	Sci	position at 2026-02-01
4867 Polites (1989 SZ)	1:55:34.64	41:05:34.90	17.8	V	Sci	position at 2026-02-01
4902 Thessandrus (1989 AN2)	11:48:00.80	1:03:26.81	17.2	V	Sci	position at 2026-02-01
5027 Androgeos (1988 BX1)	12:28:12.44	37:22:35.52	17.3	V	Sci	position at 2026-02-01
5028 Halaesus (1988 BY1)	12:55:09.31	13:14:45.14	18.0	V	Sci	position at 2026-02-01
5126 Achaemenides (1989 CH2)	12:18:03.75	27:47:05.22	18.0	V	Sci	position at 2026-02-01
5130 Ilioneus (1989 SC7)	0:32:02.69	13:48:24.37	17.7	V	Sci	position at 2026-02-01
5144 Achates (1991 XX)	2:52:21.82	26:06:42.60	17.3	V	Sci	position at 2026-02-01
5254 Ulysses (1986 VG1)	12:49:55.59	24:45:35.12	16.8	V	Sci	position at 2026-02-01
5258 Rhoeo (1989 AU1)	11:08:28.73	-2:12:42.47	17.3	V	Sci	position at 2026-02-01
5264 Telephus (1991 KC)	13:03:39.08	27:32:05.67	16.6	V	Sci	position at 2026-02-01
5284 Orsilocus (1989 CK2)	12:54:39.90	18:15:18.61	17.6	V	Sci	position at 2026-02-01
5648 Axius (1990 VU1)	3:08:11.07	12:33:55.06	17.3	V	Sci	position at 2026-02-01
5652 Amphinachus (1992 HS3)	12:11:21.33	-3:29:18.25	17.9	V	Sci	position at 2026-02-01
6090 Aulis (1989 DJ)	10:47:29.89	3:50:44.93	16.9	V	Sci	position at 2026-02-01
6545 Leitus (1986 TR6)	13:13:09.75	-21:33:41.97	18.0	V	Sci	position at 2026-02-01
7119 Hiera (1989 AV2)	11:03:36.84	-15:33:32.95	17.3	V	Sci	position at 2026-02-01
7641 Cteatus (1986 TT6)	10:43:23.25	-34:31:40.72	16.8	V	Sci	position at 2026-02-01
9799 Thronium (1996 RJ)	10:37:52.96	-29:45:35.95	17.3	V	Sci	position at 2026-02-01
11395 Iphinous (1998 XN77)	11:04:42.12	-16:41:35.81	17.6	V	Sci	position at 2026-02-01
11552 Boucolion (1993 BD4)	4:14:35.83	6:27:57.58	17.9	V	Sci	position at 2026-02-01
12126 Chersidamas (1999 RM11)	2:46:36.85	18:02:51.51	18.2	V	Sci	position at 2026-02-01
12444 Prothoon (1996 GE19)	1:58:52.60	7:32:27.97	18.0	V	Sci	position at 2026-02-01
12929 Periboea (1999 TZ1)	3:19:32.14	-12:28:09.44	17.9	V	Sci	position at 2026-02-01
14268 (2000 AK156)	10:35:04.64	-6:37:35.67	17.9	V	Sci	position at 2026-02-01
15436 Dexius (1998 VU30)	11:32:16.49	-17:13:57.16	16.6	V	Sci	position at 2026-02-01
15440 Eioneus (1998 WX4)	11:09:40.21	26:45:45.26	16.9	V	Sci	position at 2026-02-01
15502 Hypeirochus (1999 NV27)	1:18:48.18	26:02:06.98	17.9	V	Sci	position at 2026-02-01
16070 Charops (1999 RB101)	3:39:22.24	33:30:16.92	17.7	V	Sci	position at 2026-02-01
16974 Iphthime (1998 WR21)	10:57:17.21	-12:00:27.10	17.5	V	Sci	position at 2026-02-01
17492 Hippasos (1991 XG1)	4:35:40.83	15:53:35.41	17.8	V	Sci	position at 2026-02-01
20729 Opheltius (1999 XS143)	11:31:24.78	-16:58:37.58	18.1	V	Sci	position at 2026-02-01
21601 Aias (1998 XO89)	13:50:45.44	-29:57:58.89	17.8	V	Sci	position at 2026-02-01
21900 Orus (1999 VQ10)	11:44:08.18	-8:58:53.28	17.4	V	Sci	position at 2026-02-01
23135 Pheidias (2000 AN146)	12:34:05.71	-23:24:28.13	17.5	V	Sci	position at 2026-02-01
34746 Thoon (2001 QE91)	2:19:26.11	-9:19:12.53	17.7	V	Sci	position at 2026-02-01
38050 Bias (1998 VR38)	10:58:08.45	-26:35:11.81	17.4	V	Sci	position at 2026-02-01